

Job Family:	SCIENTIFIC	Reports to:	
Role Title:	Nuclear Science and Technology Fellow	Employees/ASWs managed:	0
Grade:	Grade K	Budget:	
Cost Centre:	Technology Department		

Overall Purpose:

Provide scientific direction and leadership in Nuclear Technology research domains that have either direct or indirect benefits to fusion, ensuring that state of the art scientific approaches are used in developing and delivering solutions.

As a globally acknowledged expert in nuclear fields - spanning nuclear technologies, diagnostics, radiation metrology, neutronics and nuclear data - shape the organisation's thinking in these through involvement in domestic and international nuclear programmes. Make use of the enhanced individual freedom and status of a Senior Fellow to pursue new lines of thought.

As a UKAEA Senior Fellow, be part of a cohort of acclaimed experts who will lead the scrutiny and challenge of UKAEA scientific programmes.

Accountabilities:

- Lead and direct an internationally acclaimed programme of research and innovation in nuclear technology domains.
- Ensure major personal scientific output which advances nuclear technology with demonstrable impact, comparable to that expected from a professor at an internationally leading university; produce journal publications and give international conference presentations.
- Initiate, develop and lead new experimental work areas in nuclear fields, including 14.1 MeV neutron irradiation experiments for the purposes of benchmarking radiation transport codes and nuclear data.
- Initiate and develop the field of modern neutron detection technologies at UKAEA with particular emphasis on instrumentation relevant to future fusion power.
- Lead and direct the field of applied nuclear physics & technology supporting groups across UKAEA in a range of activities for a range of customers.
- Place UKAEA at the forefront of fusion technology in the areas of neutronics, nuclear detection systems evaluating existing computational tools and develop new modelling capability required to simulate nuclear phenomena. Develop and promote novel approaches to performing nuclear analysis, including on HPC platforms.
- Design and communicate UKAEA nuclear research strategies, identifying USPs now and in the future.
- Initiate and develop new scientific methods/techniques in nuclear fields. Assesses recommendations for the development of new nuclear methods/techniques, in the context of wider organisation priorities.
- Maintain and develop strong collaborations with UK universities and overseas institutions, supervising PhD students and post-docs. Represent UKAEAs interests by acting as an expert on international panels.
- Scrutinise, challenge, and advise senior management on the UKAEA scientific strategy as part of a cohort of Senior Fellow and other acclaimed experts.

Specific Qualifications/Experience:

- Substantial experience in the nuclear physics & technology spanning fusion and related areas such as, fission, spallation, accelerator and medical domains
- High international standing, recognised by invited talks, international prizes, visiting academic/professorships, or equivalent
- Proven ability to develop research programmes and long-term vision for the scientific research area
- An established and proven track record of excellent scientific research

Additional Duties:

Manage Nuclear aspects of the Fusion Technology dept and programme
Represent UKAEA at various international meetings. Undertake high level secondment with e.g. IAEA departments and contribute expertise
Initiate and manage collaborations with various international groups and UK universities, establishing new connections to plug internal capability gaps as needed

Security Clearance:

Appointments:

Information Asset Owner

Technical/ Professional:	80%	Project Management:	5%	People Management:	15%
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Additional Role information for all jobs in this career family and grade
(This is standard information, please do not amend)